Assignment: Advanced Data Visualisation – Assessment Task 2

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1.Introduction

This report presents an in-depth visual exploration and analytical overview of the Australian Open tennis championships dataset, spanning more than 120 years of competition. The goal of this project is to uncover and communicate key insights from a high-dimensional sports dataset using Tableau, a dynamic data visualisation tool. The analysis covers player performance, win trends, gender-based comparisons, and geographical dominance.

By leveraging visual techniques such as treemaps, parallel coordinates, geographic maps, and scatter plots, the report highlights the evolution of the sport, identifies standout players, and draws comparisons across time, gender, and nationality. Special attention is given to players who have won five or more championships, offering detailed performance comparisons across sets and match conditions.

This report is designed not only to present analytical findings but also to guide the reader through a clear narrative of visual storytelling. From historical dominance and strategic consistency to global representation and performance correlations, the reader will observe how effective visualisations can reveal layers of meaning within complex datasets.

2. Dataset Summary

The Australian Open dataset provides over a century of final match records from both men's and women's tournaments. Each row in the dataset represents a championship final, while each column captures a key match or player attribute, including outcomes, nationalities, seeds, and scores.

- Key Variables and Formats:
- Year (numeric): Indicates the tournament year.
- Gender (categorical): Men's or Women's final.
- Champion / Runner-up (text): Names of the winner and finalist.
- Nationalities (text): Country representation of both players.
- Seeds (numeric): Tournament seeding of champion and runner-up.
- Match Duration ("Mins") (numeric, often missing): Excluded from visualisation due to incompleteness.
- Set Scores (structured numeric): Ten fields (e.g., 1st-Won, 1st-Loss, ..., 5th-Won, 5th-Loss) representing match outcomes per set.

Data Characteristics:

Covers over 120 years of finals across both genders.

Set scores reveal that many matches conclude in under five sets, leaving later columns blank—an expected structural pattern.

The dataset required no major cleaning, aside from omitting the “Mins” column from analysis.

The clean, well-organised tabular structure makes the data ideal for Tableau visualisation, particularly for treemaps, scatter plots, and parallel coordinates.

Support for Visualisation Techniques:

Treemaps were used to compare the top-performing players who have won five or more titles. The area and colour intensity allowed for quick visual comparison of title counts, player gender, and nationality, highlighting historical dominance.

Parallel coordinates visualised consistency across set win rates (1st to 5th sets) for different champions, allowing multi-dimensional analysis of performance patterns in match sets.

Geographic maps (filled maps) illustrated the global distribution of champions by country. The shading gradient made it easy to identify national trends in tennis dominance over the history of the tournament.

Scatter plots (by gender) were used to analyse the relationship between average games won and win rate across men's and women's finals. This revealed performance clusters and potential differences between male and female champions.

Scatter plots (over time) were also used to analyse average win rate changes over time, plotted against the year of victory. This enabled temporal trend analysis and highlighted shifts in performance consistency across decades.

3. Data Preparation and Transformation

Although the dataset was mostly structured and analysis-ready, several essential transformations were applied to support accurate and insightful visualisation using Tableau.

Win Rate Calculations
Using the structured set-level columns, multiple win-related metrics were derived to enable performance analysis:
Set Win Rate:
For each individual set (1st to 5th), the following formula was used:

$$Win\ Rate\ =\ Games\ Won\ /\ (Games\ Won\ +\ Games\ Lost)$$

The calculation was conditionally applied only when both values were available to avoid errors from missing or zero values.

$$= IF(AND(ISNUMBER(Won), ISNUMBER(Loss), (Won + Loss) <> 0), Won/(Won + Loss), "")$$

Total Games Won and Played:
New fields were created to calculate the total number of games won and total games played by each champion per final.

Overall Win Rate:
An aggregate win rate per match was calculated as:

$$Total\ Win\ Rate\ =\ Total\ Games\ Won\ /\ Total\ Games\ Played$$

This metric was critical for generating scatter plots and performance trend visualisations.

Filtering Top Champions
To focus on historically dominant players, a filter was applied to isolate champions with five or more titles. This subset was used to produce the treemap, providing a high-level visual comparison of the most successful players.

Handling Missing or Incomplete Data

- "Walkover" Matches:
  Finals such as the 1966 Women's match that ended via walkover were retained for historical completeness but excluded from charts involving performance metrics due to the absence of valid match data.

- Incomplete Duration Data ("Mins"):

The "Mins" column was largely incomplete and excluded from all visualisations to maintain consistency and reliability.

Preparation for Visualisation
All calculated fields were created in Excel using formulas and validated before being imported into Tableau. Field names were clearly labelled (e.g., “1st Set Win Rate”, “Total Games Won”) to support seamless integration into Tableau dashboards and ensure clarity across all visualisations.

These calculated metrics formed the foundation for advanced visualisations such as scatter plots, parallel coordinates, and nationality comparisons, and were essential in crafting a coherent and data-driven narrative throughout the report.

4. Visual Analysis and Findings

Each visualisation was carefully designed to address key themes in the dataset: national dominance, gender-based trends, match-level consistency, long-term evolution, and individual excellence. Together, they form a cohesive narrative that enables exploration of both macro-level trends and player-specific performance.

4.1 Geographic Map – National Representation of Champions.

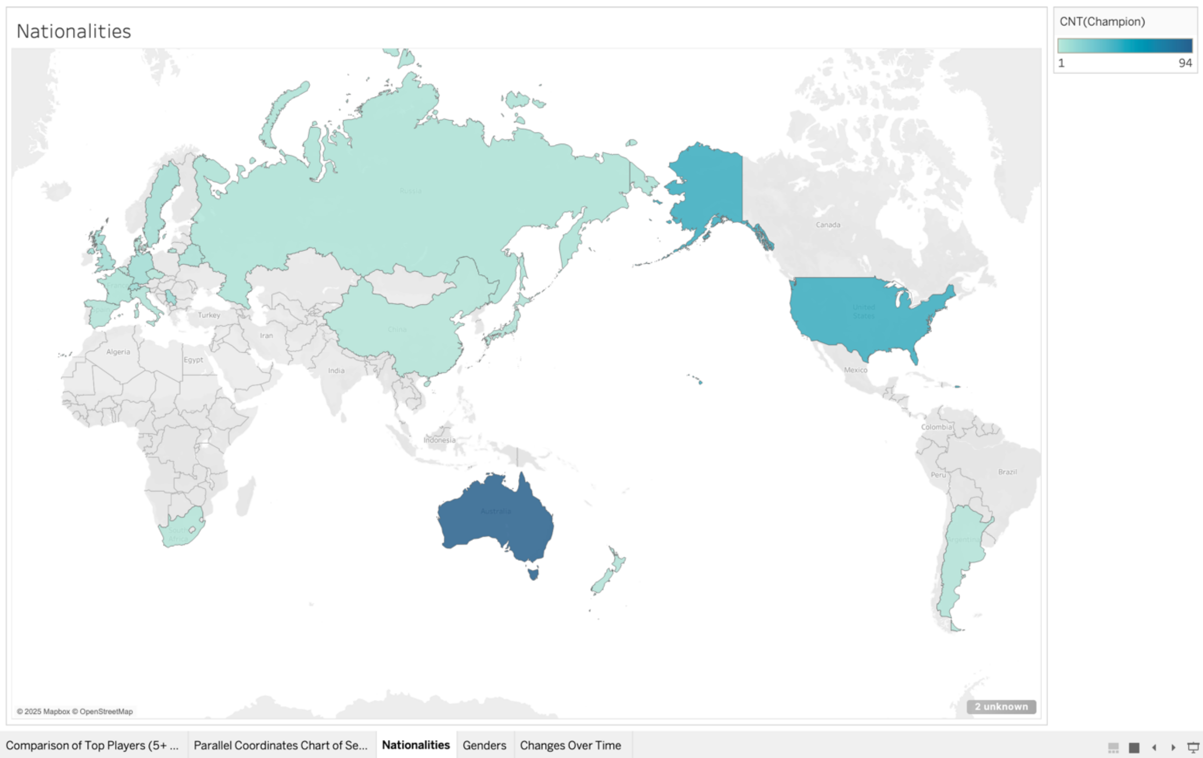

Figure 1:National Representation of Champions.

The geographic map in **Figure 1** visualises the distribution of Australian Open champions by nationality, using a filled map format where darker shades indicate a higher number of titles won by players from a given country. This visualisation enables a quick comparative understanding of global tennis dominance throughout the tournament's history.

As the chart illustrates, **Australia** stands out prominently in the darkest blue, reflecting its historical dominance with the highest count of champions. This trend is closely tied to the tournament's origin and its early participant pool, which heavily favoured local players in its formative decades. The **United States** also features strongly, indicating its sustained impact, particularly during the Open Era with players such as Serena Williams and Andre Agassi. Other countries with noticeable contributions include **Switzerland**, **Germany**, and **Serbia**, highlighting the broader internationalisation of the sport in recent years.

In contrast, large regions such as Africa, parts of Asia, and South America appear unshaded, signalling limited or no representation among champions. This absence reflects global disparities in access to elite tennis training infrastructure and competitive exposure.

The colour gradient used in this map helps the viewer grasp relative success at a national level without relying on explicit counts. The map succeeds in delivering an intuitive overview of **regional tennis excellence**, and complements player-level analysis by anchoring it to broader national trends.

4.2 Scatter Plot – Gender-Based Game and Win Rate Distribution.

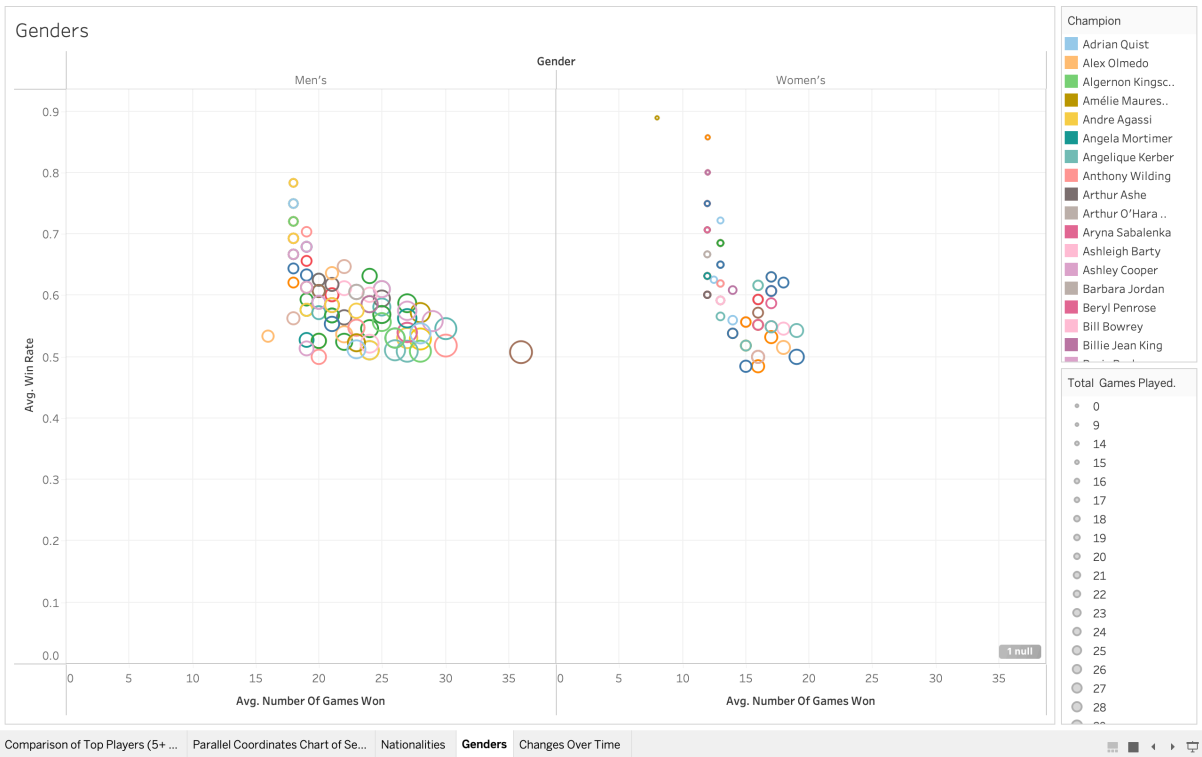

Figure 2:Gender-Based Game and Win Rate Distribution.

The scatter plot in **Figure 2** compares male and female Australian Open champions by visualising the relationship between the **average number of games won per final** (x-axis) and the **average win rate** (y-axis). The data is segmented by gender, providing two distinct distributions within the same visual space. Each point represents a player, and the size of the bubble reflects the total number of matches played.

In the men's panel, the data points display a **broader spread**, particularly in the number of games won, which reflects the best-of-five format used in men's finals. This leads to higher total game counts and greater variability in performance. In contrast, the women's distribution is **more compact and concentrated**, due to the best-of-three format, which results in more consistent average game counts.

Notably, the **average win rates among women tend to be slightly higher** than their male counterparts. This may suggest that female champions often win in a more dominant fashion or encounter less variability in match length. Meanwhile, several male players show moderate win rates despite high game counts, indicating longer or more contested matches.

The bubble sizes add a third dimension to the analysis, helping identify players with extensive match experience. For example, larger circles in both groups denote champions who consistently performed across many finals, adding context to their efficiency or endurance.

By separating gender while preserving the same axes, this chart facilitates a **side-by-side performance comparison** that is both visually intuitive and analytically rich. It supports analysis of match structure differences, competitive intensity, and consistency within each category.

The plot effectively reveals gender-based patterns in win behaviour and contributes to a deeper understanding of performance dynamics in the Australian Open across both men's and women's championships.

4.3 Parallel Coordinates – Set Win Rate Consistency.

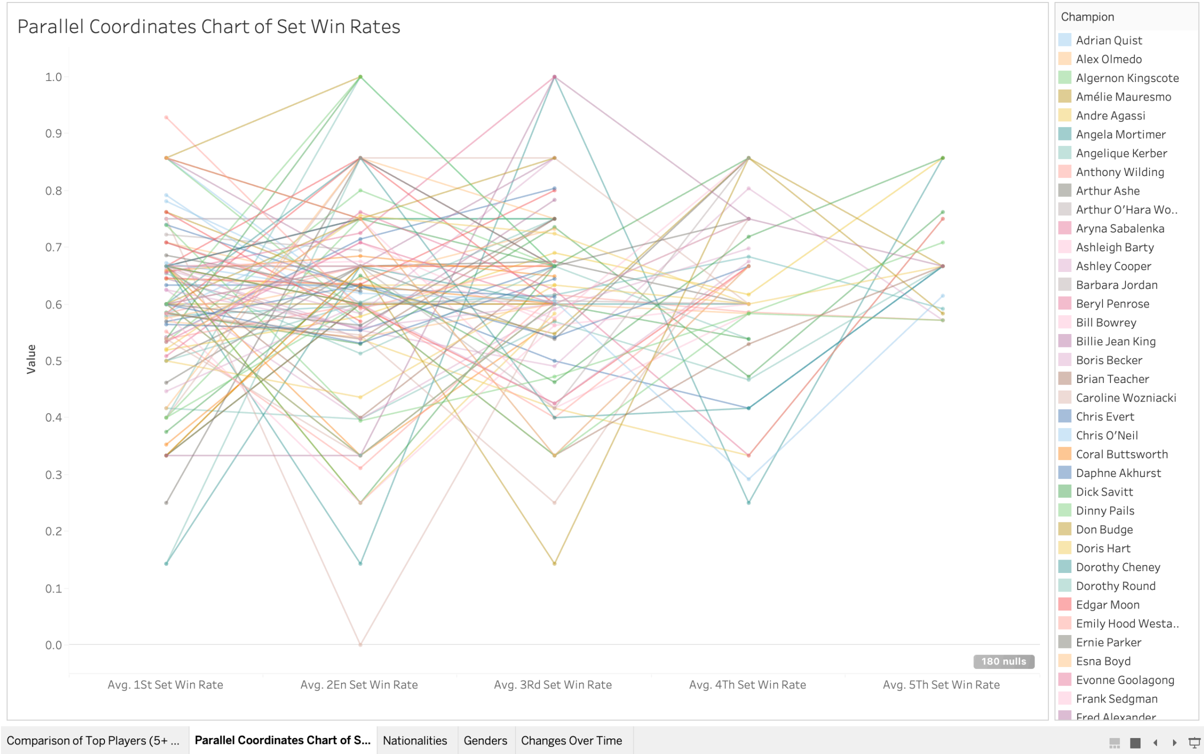

Figure 3:Parallel Coordinates – Set Win Rate Consistency.

The parallel coordinates visualisation in **Figure 3** presents the **set-level win rates** of champions across the 1st to 5th sets, offering a multidimensional view of match consistency and performance trends. Each line represents a different champion, with the vertical axes corresponding to the average win rate for each set. The values range from 0 (no games won) to 1 (perfect win rate), allowing clear differentiation between dominant and inconsistent performances.

A key insight from this chart is the **relative stability of win rates across the first three sets** for most champions, suggesting strong early-match strategies and control. However, **noticeable dips in the 4th and 5th sets** are observed in multiple lines, indicating that prolonged matches may challenge a player's endurance or strategic adaptability. In contrast, some players maintain **high and stable win rates across all five sets**, revealing their capacity to sustain performance under pressure.

This visual is particularly powerful for identifying subtle patterns that are not easily observed in aggregated statistics. For example, it becomes apparent which champions tend to dominate early and which maintain consistent superiority across longer matches. The variation in slopes also reveals **potential vulnerabilities**, such as players who start strong but drop off in later sets.

By enabling a layered analysis of match progression, this chart adds depth to the evaluation of player performance, highlighting **not just who wins**, but **how they win across different stages of the match**. It is an effective tool for comparing champions with differing match styles, endurance levels, and strategic profiles.

4.4 Scatter Plot – Temporal Evolution of Win Rates.

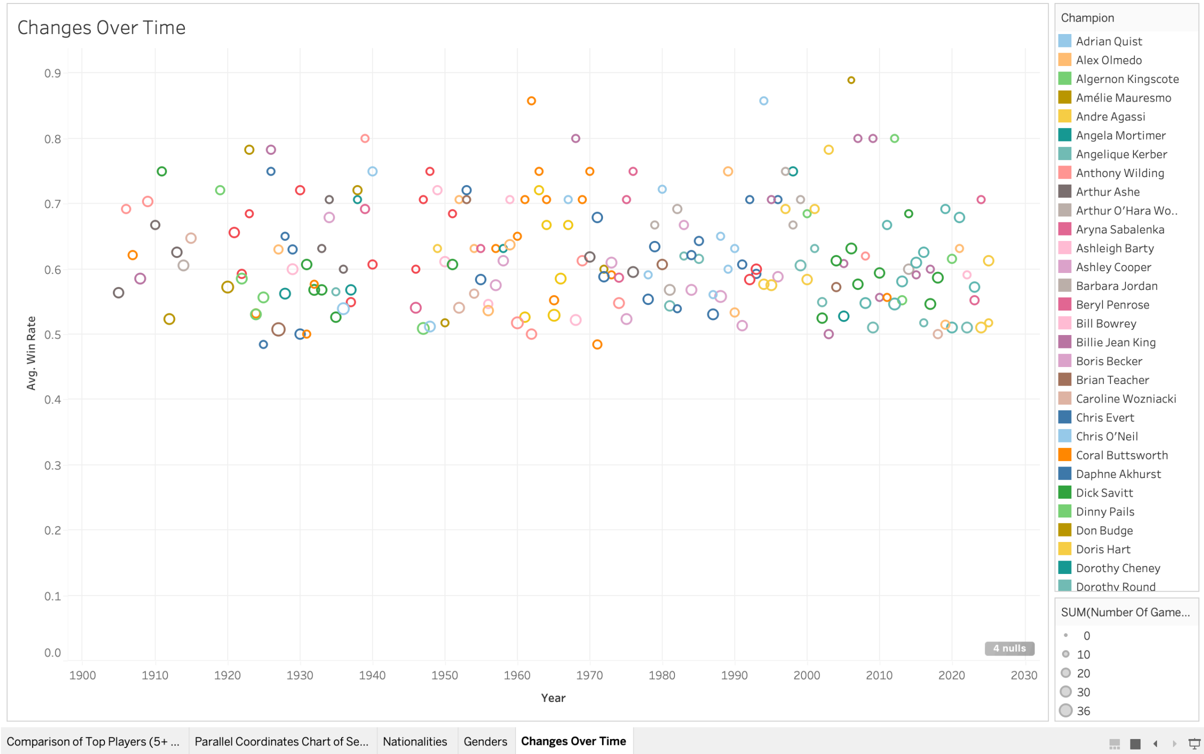

Figure 4:Temporal Evolution of Win Rates.

The scatter plot in **Figure 4** explores the evolution of performance at the Australian Open by plotting the **average win rate of champions** against the **year of the tournament**. Each point represents a champion's final match win rate in a given year, with colours used to distinguish between individual players, and bubble size representing the number of games won.

The visual reveals a **consistently clustered range of win rates** between approximately 0.55 and 0.75 throughout the 120-year span of the tournament. This consistency suggests that despite changes in era, technology, and playing conditions, the level of dominance required to win a final has remained relatively stable. However, there are subtle signs of an **upward trend in win rates in recent decades**, which could be attributed to advancements in player fitness, data-driven coaching, and improved match preparation.

Some early 20th-century data points appear as **outliers**, particularly those with unusually high or low win rates. These may reflect inconsistencies in historical match structures or limitations in available data. For example, walkovers or finals affected by injury may distort the trend, especially in earlier years.

This chart provides a powerful longitudinal view of competitive intensity in finals. It allows readers to trace **how the average level of control exerted by champions has shifted (or remained stable)** over time. It also contextualises modern champions within the broader history of the tournament, offering both a **temporal benchmark** and an intuitive snapshot of change.

By combining performance data with a historical timeline, this visual delivers a compelling narrative of evolution in professional tennis, framed around the Australian Open's most elite competitors.

4.5 Top Champions (5+ Titles).

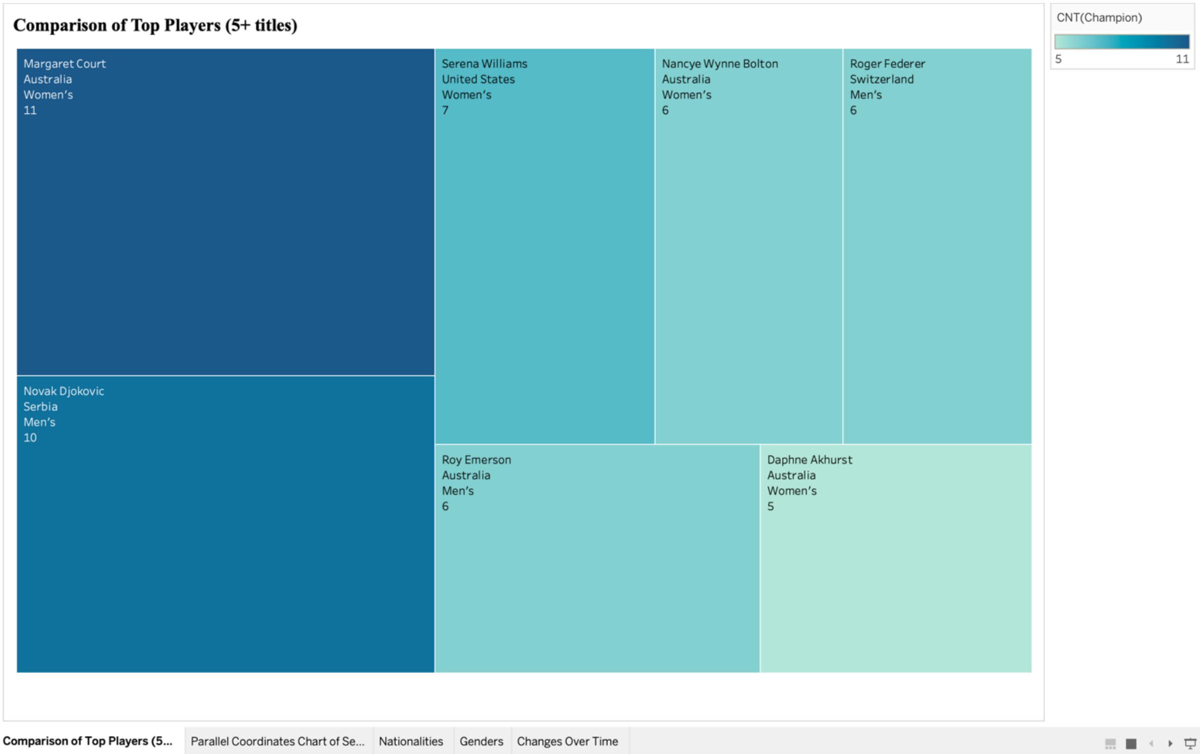

Figure 5: Champions with 5+ Wins

The treemap visualisation in **Figure 5** highlights the top-performing players in the history of the Australian Open who have won **five or more titles**. Each rectangle represents a champion, with the size of the tile indicating the number of titles won, and the colour intensity—ranging from light turquoise to deep blue—serving as a visual indicator of relative dominance. At the top of the spectrum, **Margaret Court** appears in the darkest shade, reflecting her record of **11 titles**, the most in tournament history. She is followed by **Novak Djokovic** (10 titles) and **Serena Williams** (7 titles), both displayed in slightly lighter hues, denoting their elite but comparatively fewer achievements. Other prominent names such as **Roger Federer**, **Roy Emerson**, and **Nancye Wynne Bolton** appear in progressively lighter shades, while **Daphne Akhurst**, with 5 titles, is represented in one of the lightest tones.

This visual approach prioritises comparative storytelling over raw numerical detail, allowing viewers to quickly identify the most dominant figures without needing to examine specific values. Notably, the chart includes both male and female players in a unified layout, promoting **gender-neutral performance analysis** based purely on achievement. The visual clustering of several female champions in the darker, larger sections also underscores their significant contributions, particularly from Australia in the early to mid-20th century. By offering a concise yet visually engaging summary of top-level performance, the treemap serves as an effective entry point into the dataset, framing the broader narrative of dominance and excellence at the Australian Open.

5. Advantages and Disadvantages of Tableau

Tableau offers several advantages that make it a powerful tool for data visualisation and storytelling. Its user-friendly drag-and-drop interface allows users to create complex graphics with ease, while its wide range of chart types and formatting options supports highly customisable dashboards. The ability to connect with various data sources and apply real-time filtering enhances its utility for both static analysis and interactive exploration. For students and educators, Tableau provides free academic licenses, making it an accessible tool in educational environments.

However, Tableau does come with limitations. It lacks advanced statistical and machine learning capabilities compared to platforms like R or Python, making it less suitable for in-depth predictive modelling. Performance may also be affected when dealing with very large datasets, especially without data extracts. Additionally, while simple visuals are easy to produce, mastering calculated fields and complex dashboards requires a steeper learning curve.

7. Conclusion

This report has delivered a comprehensive visual analysis of over a century of Australian Open championship data, uncovering key insights through a combination of advanced data preparation and effective visual storytelling using Tableau. By focusing on five core visualisations—treemap, parallel coordinates, geographic map, and two scatter plots—the project explored national dominance, gender-based performance differences, consistency across match sets, and the evolution of win rates over time.

Through the integration of calculated fields such as set-level win rates, total games won, and title counts, the analysis moved beyond surface-level statistics to reveal deeper performance patterns. The treemap highlighted the legacy of top champions, the parallel coordinates captured strategic match control, and the geographic map contextualised success on a global scale. The scatter plots facilitated nuanced gender-based comparison and historical performance tracking, adding dimension to the understanding of how tennis has changed and persisted.

Tableau's role was instrumental in transforming a complex, high-dimensional dataset into accessible, engaging visual formats. Its interactivity, clarity, and flexibility enabled the creation of visuals that not only inform but also narrate—bridging data and interpretation seamlessly.

In summary, the project not only achieved its objective of identifying meaningful trends within the Australian Open data, but also demonstrated the power of visual analytics to drive clarity, engagement, and insight in sports data analysis